A Simplified Model of Serpentine Locomotion

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Note. This document is an updated presentation of the dynamical modeling and control components of my 2017 MS thesis and paper [Racioppo and Ben-Tzvi, 2019].

Abstract

We revisit classical models of serpentine locomotion and show that, under appropriate assumptions, the dynamics admit a low-dimensional approximation governing the center-of-mass motion. While full models require solving constrained differential-algebraic systems, we demonstrate that, when the body follows a prescribed traveling-wave gait, the resulting motion can be approximated by a single first-order equation for the center-of-mass velocity. This reduction is obtained by combining a representative-link approximation with time-scale separation and period averaging. We identify the key mechanism underlying propulsion as a phase offset between lateral motion and body orientation, induced by anisotropic friction. In addition, we analyze the structure of the phase-offset optimization problem and derive a closed-form approximation for the optimal joint phase offset. Together, these results provide (i) a simplified dynamical model that predicts steady-state velocity, (ii) a principled characterization of optimal phase relationships, and (iii) an interpretation of serpentine motion in terms of traveling-wave structure and phase interactions. This perspective yields a unified and computationally efficient framework for understanding serpentine locomotion, connecting gait design, physical modeling, and control.

1 Introduction

Serpentine locomotion, or lateral undulation, is the dominant mode of motion in biological snakes and has long served as a template for snake robot design. It consists of a traveling transverse wave that propagates along the body. Unlike standing waves, this motion relies on a coordinated phase relationship between body segments: each segment pushes laterally against the ground while the overall wave advances from head to tail. When coupled with anisotropic friction—higher resistance normal to the body than along it—this lateral motion is converted into forward propulsion.

In robotic implementations, this continuous body deformation is approximated by a finite set of rigid links connected by actuated joints. The standard parameterization of planar serpentine motion prescribes the joint angles as a traveling sinusoidal wave:

$$\theta_{\mathrm{ref},i} = \alpha \sin(\omega t + (i-1)\delta) + \phi_0, \tag{1}$$

for $i \in \{1, \dots, N-1\}$. Here, α is the amplitude of angular displacement, ω is the temporal frequency, and δ is the phase offset between adjacent joints. The parameter ϕ_0 induces a constant curvature and enables turning.

While this representation is typically viewed as a high-dimensional control input over $N-1$ joints, it is in fact governed by a small set of parameters $(\alpha, \omega, \delta, \phi_0)$ that define a traveling wave along the body. In this sense, the full joint motion is determined by a low-dimensional parameterization of the waveform.

Despite this structure, most existing models of snake locomotion operate at the level of full multibody dynamics, requiring the solution of large systems of coupled differential-algebraic equations. While these models are physically accurate, they often make it difficult to isolate the mechanisms that govern locomotion performance, such as the role of phase relationships and anisotropic friction.

In this work, we develop two complementary perspectives on serpentine locomotion. First, we show that, when the body motion is prescribed as a traveling-wave gait and appropriate time-scale separation holds, the center-of-mass dynamics can be approximated by a low-dimensional model. By treating the joint motion as a controlled input and averaging the resulting frictional forces, we derive a reduced description in which forward propulsion is governed by a phase offset between lateral motion and body orientation, and the resulting behavior is captured by a single first-order equation for the center-of-mass velocity.

Second, we analyze the structure of the phase-offset optimization problem and derive a closed-form approximation for the optimal joint phase offset. This analysis reveals how propulsion depends on a structured aggregation of phase interactions along the body, providing a complementary perspective on the role of the traveling-wave parameters.

By making these structures explicit, we aim to bridge the gap between detailed physical models and intuitive, interpretable descriptions of locomotion, providing tools that are both analytically tractable and physically grounded.

2 Related Work

Early work in snake robotics is largely attributed to Hirose [1993], who introduced the serpenoid curve as a mathematical description of serpentine locomotion. This formulation models the body of a snake as a traveling sinusoidal wave and remains the foundation for most subsequent work on snake robot kinematics.

Extensions of this idea have focused on alternative curvature parameterizations and their effect on locomotion efficiency. Ma [1999] analyzed different curvature profiles and compared their performance, while Chernousko [2005] proposed a complementary approach in which locomotion is constructed from elementary joint motions in low-degree-of-freedom systems. These works highlight that a wide range of locomotion patterns can be generated from relatively simple underlying structures.

A significant body of work has also focused on control and physical modeling. Liljebäck et al. [2012] and Liljebäck and Pettersen [2011] developed path-following and waypoint guidance controllers for snake robots executing serpenoid-like motions. In parallel, Liljebäck et al. [2010] established that anisotropic ground friction is a necessary condition for controllable serpentine locomotion, formalizing the role of directional frictional forces in generating propulsion. Further analysis of controllability and stability was provided by Liljebäck et al. [2011].

More comprehensive treatments of snake robot modeling and locomotion can be found in the survey by Transeth et al. [2009] and the monograph by Liljebäck et al. [2013], which cover both kinematic and dynamic approaches. Earlier experimental and control-oriented studies, such as Sato et al. [2002], also contributed to the development of practical locomotion strategies.

In addition to modeling and control, optimization of gait and design parameters has been widely studied, particularly with the goal of maximizing forward speed. This includes tuning link lengths, mass distributions, and control parameters to improve performance [Hirose, 1993, Chernousko, 2005].

Prior work has primarily approached serpentine locomotion from two complementary perspectives: kinematic descriptions based on traveling waves (e.g., the serpenoid curve), and high-fidelity dynamic models that capture frictional interactions and multibody effects. While both perspectives have led to significant advances, they often remain disconnected: kinematic models provide intuition but lack predictive power, whereas dynamic models are accurate but difficult to analyze and interpret.

This work aims to bridge this gap by making explicit the low-dimensional structure underlying serpentine locomotion. We show that the traveling-wave parameterization used in kinematic models induces a corresponding simplification in the dynamics, allowing the high-dimensional multibody system to be approximated by a single first-order equation governing center-of-mass motion. In this view, key quantities such as the phase offset and friction anisotropy emerge as the primary determinants of locomotion performance, providing a unified interpretation that connects gait design, physical modeling, and control.

3 Multibody Dynamic Model

We present a multibody dynamic (MBD) model of a snake robot, which serves as a high-fidelity reference for the simplified model developed in the subsequent section. Unlike reduced models that impose no-slip constraints, this formulation explicitly models frictional forces and internal constraint forces via Lagrange multipliers.

3.1 Kinematics and Constraints

The snake robot is modeled as N rigid links connected by revolute joints. Each link i has half-length L_i, mass m_i, and moment of inertia I_i. The generalized coordinates are:

$$q = [q_1, \ldots, q_N]^T, \quad q_i = [r_i, \phi_i]^T$$

where $r_i = [x_i, y_i]^T$ is the position of the center of mass and ϕ_i is the orientation.

Revolute joints impose $2(N-1)$ algebraic constraints, collected in:

$$\Phi(q, t) = 0.$$

The rotation matrix mapping global to local coordinates is:

$$R_i = \begin{bmatrix} \cos\phi_i & -\sin\phi_i \\ \sin\phi_i & \cos\phi_i \end{bmatrix}.$$

3.2 Friction Model

Friction is modeled using anisotropic coefficients c_n (normal) and c_t (tangential). The viscous friction force acting on link i is:

$$F_{f,i}(r_i, \phi_i) = -R_i^T \Omega_i R_i \dot{r}_i, \tag{2}$$

where $\Omega_i = \mathrm{diag}(c_{n,i}, c_{t,i})$.

Rotational friction is computed by approximating each half-link as n discrete masses with spacing:

$$l_{n,i} = \frac{2L_i}{2n+1}.$$

For angular velocity $\dot{\phi}_i$, segment velocities are:

$$v_{x_{n,i}} = \dot{\phi}_i l_{n,i} \cos\phi_i, \tag{3}$$

$$v_{y_{n,i}} = -\dot{\phi}_i l_{n,i} \sin\phi_i. \tag{4}$$

In the limit $n \to \infty$, this yields an equivalent force acting at:

$$l'_{\mathrm{eff},i} = \frac{1}{2}.$$

Defining:

$$v_{\mathrm{eff},i} = \begin{bmatrix} \dot{\phi}_i l'_{\mathrm{eff},i} \cos\phi_i \\ \dot{\phi}_i l'_{\mathrm{eff},i} \sin\phi_i \end{bmatrix},$$

the frictional torque is:

$$\tau_{f,i}(\phi_i) = l'_{\mathrm{eff},i} F_{f,i}(v_{\mathrm{eff},i}). \tag{5}$$

3.3 Equations of Motion

Let Q_A denote the generalized force vector and define:

$$M = \mathrm{diag}(m_1, m_1, I_1, \ldots, m_N, m_N, I_N).$$

The constrained variational equations are:

$$\delta q^T (M\ddot{q} - Q_A) = 0. \tag{6}$$

Introducing Lagrange multipliers λ, the system becomes the DAE:

$$\begin{bmatrix} M & \Phi_q^T & 0 \\ \Phi_q & 0 & 0 \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Q_A \\ \gamma \\ \dot{q} \end{bmatrix}, \tag{7}$$

where Φ_q is the constraint Jacobian and γ contains acceleration-independent terms.

Initial conditions must satisfy:

$$\Phi_q \dot{q} = -\Phi_t.$$

The system state

$$Y = [\dot{q}^T \;\; \lambda^T \;\; q^T]^T$$

is obtained using a numerical DAE solver.

4 Partial Feedback Linearization of the Joint Dynamics

An N-link snake robot moving in the plane can be modeled using a multibody dynamic approach as a serial chain of straight links joined by revolute joints. The system's dynamics are described by $3N$ differential algebraic equations (DAEs) for the links, complemented by $2(N-1)$ algebraic constraint equations for the revolute joints.

The generalized coordinates for the i-th link's center of mass (CM) are specified by a vector $q_i = [r_i, \phi_i]^T$, where $r_i = [x_i, y_i]^T$ denotes the global position in the xy-plane, and ϕ_i is its global orientation. Introducing a vector of Lagrange multipliers $\lambda_i = [\lambda_{x,i}, \lambda_{y,i}]^T$ to enforce the joint constraints, a diagonal 3×3 generalized mass matrix M_i, a generalized external force vector $F_{\mathrm{ext},i}$, and a vector of generalized actuation forces U_i for the i-th link, the system's equations of motion can be written as:

$$\begin{cases} M_i \ddot{q}_i = F_{\mathrm{ext},i} + f_i(q, \dot{q}) + \lambda_i + U_i \\ \Phi_j(q, t) = 0 \end{cases} \tag{8}$$

Here, $f_i(q, \dot{q})$ represents internal forces and torques (e.g., Coriolis and centrifugal terms), while $\Phi_j(q, t) = 0$ represents the j-th revolute joint constraint equation. Since joint torques are the only control inputs, $U_i = [0, 0, \tau_i]^T$.

The goal of feedback linearization is to transform a nonlinear system into a linear one by canceling nonlinear terms with appropriate control inputs. For a snake robot, the angular dynamics are of primary interest. The control input can be decomposed as:

$$U_i = U_{\mathrm{lin},i} + \bar{U}_i \tag{9}$$

The term $\bar{U}_i$ tracks a desired angular trajectory, while $U_{\mathrm{lin},i}$ cancels nonlinear dynamics.

The full system can be written as:

$$M\ddot{q} = F_{\mathrm{ext}}(q, \dot{q}) + F_{\mathrm{int}}(q, \dot{q}) + \Phi_q^T \lambda + U \tag{10}$$

The angular component for link i is:

$$I_i\ddot{\phi}_i = (F_{\text{ext},i})_\phi + (F_{\text{int},i})_\phi + (\Phi_q^T\lambda)_i + \tau_i \tag{11}$$

Setting:

$$\tau_i = -(F_{\text{ext},i})_\phi - (F_{\text{int},i})_\phi - (\Phi_q^T\lambda)_i + \bar{U}_i$$

yields:

$$I_i\ddot{\phi}_i = \bar{U}_i \tag{12}$$

The DAE system is:

$$\begin{bmatrix} M & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Q_A \\ \gamma \end{bmatrix} \tag{13}$$

Expanding:

$$M\ddot{q} + \Phi_q^T\lambda = Q_A \tag{14}$$

$$\Phi_q\ddot{q} = \gamma \tag{15}$$

Solving:

$$\ddot{q} = M^{-1}(Q_A - \Phi_q^T\lambda) \tag{16}$$

$$\Phi_q M^{-1}(Q_A - \Phi_q^T\lambda) = \gamma \tag{17}$$

$$\Phi_q M^{-1}Q_A - \Phi_q M^{-1}\Phi_q^T\lambda = \gamma \tag{18}$$

Thus:

$$\Phi_q M^{-1}\Phi_q^T\lambda = \Phi_q M^{-1}Q_A - \gamma \tag{19}$$

Let $A = \Phi_q M^{-1}\Phi_q^T$. Then:

$$\lambda = A^{-1}(\Phi_q M^{-1}Q_A - \gamma) \tag{20}$$

Substituting λ back yields the linearizing control.

A key feature of this formulation is that feedback linearization is performed directly on the differential-algebraic system, rather than after reducing it to an ODE via coordinate elimination. By retaining the constraint equations and explicitly solving for the Lagrange multipliers λ, the controller has direct access to the internal constraint forces.

This makes it possible to cancel the contribution of $(\Phi_q^T\lambda)$ within the control law, rendering the angular acceleration of each joint locally decoupled from the rest of the chain.

Working in the DAE framework also avoids the need to form and manipulate the full joint-space mass matrix, whose complexity grows rapidly with the number of links. Instead, the structure of the constraint equations is exploited directly, yielding a computationally efficient and exact linearization of the angular dynamics.

4.1 Traveling-Wave Control Law

Under the feedback-linearized dynamics, the joint angles can be driven to track a prescribed trajectory. In this work, we consider the standard traveling-wave parameterization:

$$\theta_{\text{ref},i} = \alpha\sin(\omega t + (i-1)\delta) + \phi_0.$$

This control law enforces a coordinated wave of curvature along the body. The phase offset δ determines the spatial structure of the wave, while ω sets its temporal frequency. As shown in the following section, the interaction between this imposed kinematic pattern and anisotropic friction gives rise to forward propulsion. Forward propulsion is governed primarily by the phase offset δ in combination with anisotropic friction. Setting $\phi_0 = 0$ produces straight-line motion, while nonzero ϕ_0 induces turning by introducing a constant curvature.

5 Derivation of an Approximation for the Optimal Joint Phase Offset

Liljebäck et al. Liljebäck et al. [2013] showed that δ_{opt}, the δ that maximizes the speed of a snake robot (in a simplified dynamic model) during serpentine locomotion, is given by the δ that maximizes

$$k_\delta = \sum_{i=1}^{N-1} \sum_{j=1}^{N-1} c_{ij} \sin\big((j-i)\delta\big), \tag{21}$$

where c_{ij} is the ijth entry of $C = A\tilde{D} \in \mathbb{R}^{(N-1)\times(N-1)}$, with $\tilde{D} = D^T(DD^T)^{-1} \in \mathbb{R}^{N\times(N-1)}$, and

$$A = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & -1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(N-1)\times N},$$

the so-called addition and difference matrices.

We obtain a closed-form approximation of δ as a function of N by deriving some properties of $C = AD^T(DD^T)^{-1}$. We define $n = N-1$ to clean up the notation, and

$$B = (DD^T)^{-1} \in \mathbb{R}^{n\times n},$$

with ijth element b_{ij}. Since a matrix times its transpose is symmetric, and the inverse of a symmetric matrix is also symmetric, B is symmetric.

Proposition 1. The sum of the kth diagonal of $B \in \mathbb{R}^{n\times n}$ is

$$b_k \equiv \frac{T_k}{n+1},$$

where T_k is the kth tetrahedral number $T_k = \frac{1}{6}k(k+1)(k+2)$.

By the kth diagonal we mean counting from the bottom-left or top-right corner of the matrix and moving toward the main diagonal. For example, a B matrix of dimension $n = 4$, with each diagonal marked with its own bracket style, is:

$$B_4 = \frac{1}{5}\begin{bmatrix} 4 & \{3\} & [2] & (1) \\ \{3\} & 6 & \{4\} & [2] \\ [2] & \{4\} & 6 & \{3\} \\ (1) & [2] & \{3\} & 4 \end{bmatrix}.$$

To prove Proposition 1, we establish three properties.

(1.1) The lower-left element and the top-right element of B are both $\frac{1}{n+1}$, and the entries increase sequentially in units of $\frac{1}{n+1}$ as we move left along the top row, right along the bottom row, up along the leftmost column, or down along the rightmost column.

Proof of (1.1). For the first column of B, the relation $DD^TB = I$ gives:

$$-b_{n-1,1} + 2b_{n,1} = 0 \;\Rightarrow\; b_{n-1,1} = 2b_{n,1},$$

$$-b_{n-2,1} + 2b_{n-1,1} - b_{n,1} = 0 \;\Rightarrow\; b_{n-2,1} = 3b_{n,1}.$$

That the first column increases sequentially by factors of $b_{n,1}$ from the nth element to the first can be proved by induction. Suppose $b_{n-i,1} = (i+1)b_{n,1}$ for some $i \geq 2$. Then

$$-b_{n-i-1,1} + 2b_{n-i,1} - b_{n-i+1,1} = 0 \;\Rightarrow\; b_{n-i-1,1} = 2(i+1)b_{n,1} - ib_{n,1} = (i+2)b_{n,1},$$

which proves the inductive step. Reaching the first element, we have

$$2b_{1,1} - b_{2,1} = 1 \;\Rightarrow\; 2nb_{n,1} - (n-1)b_{n,1} = 1 \;\Rightarrow\; b_{n,1} = \frac{1}{n+1}.$$

Applying the same argument to the last column of B shows that its elements increase in units of $\frac{1}{n+1}$ from the top-right corner to the lower-right corner. Since B is symmetric, the top and bottom rows increase in the same way. □

(1.2) The first n diagonals (starting at the bottom-left corner) of a B matrix of dimension n are precisely the diagonals of all of its lower-dimensional counterparts, scaled by $\frac{1}{n+1}$.

Proof of (1.2). The ith element on the kth diagonal, indexed as $b_{n-(k-1)+i,\,k+i}$ for $k \in [1, n]$ and $i \in [1, k]$, is recursively generated by the two elements directly below it:

$$-b_{n-(k-1)+i,\,k+i} + 2b_{n-(k-1)+i+1,\,k+i} - b_{n-(k-1)+i+2,\,k+i} = 0,$$

until we reach the second-to-bottom row, where $b_{n-(k-1)+i,\,k+i} = 2b_{n-(k-1)+i+1,\,k+i}$. Thus the first n diagonals of B are generated from the bottom row in the same manner for a B matrix of any dimension, differing only by the bottom row's scaling factor of $\frac{1}{n+1}$. □

(1.3) The sum of the main diagonal of B is $b_n = \frac{T_n}{n+1}$.

Proof of (1.3). The matrix

$$DD^T = \begin{bmatrix} 2 & -1 & & 0 \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ 0 & & -1 & 2 \end{bmatrix}$$

is the so-called second difference matrix, in which the sum of the kth lower diagonal (moving from the bottom-left toward the main diagonal) is

$$2\sum_{i=1}^{k} b_{i+(n-k),i} - \left(\sum_{i=1}^{k-1} b_{i+(n-k),(i-1)} + \sum_{i=1}^{k-1} b_{i+(n-k),(i+1)}\right).$$

Since $DD^TB = I \in \mathbb{R}^{n\times n}$, we have $\mathrm{tr}(DD^TB) = n$, and since $(DD^T)^{-1}$ is symmetric, we can write

$$\sum_{i=1}^{n} b_{i,i} - \sum_{i=1}^{n-1} b_{i,i-1} = 2$$

for the main diagonal. Define $f(k)$ as the sum of the kth diagonal in an n-dimensional B matrix, moving from the outside in. For the case $k = n$, the recursion gives

$$f(n) - f(n-1) = n = \frac{n(n+1)}{2} \cdot \frac{1}{n+1} \cdot 2,$$

which we recognize as $\frac{T_n}{n+1} \cdot \frac{n+1}{1}$.

We can now prove by inverse induction on n that the kth diagonal of a B matrix of dimension n sums to $\frac{T_k}{n+1}$. This is clearly the case for $n = 1$ or $n = 2$. For $n = m \geq 2$, the sum of the main diagonal is $\frac{T_m}{m+1}$, and the sums of the outer diagonals are all given by $\frac{T_k}{m+1}$, provided that the claim holds for all $n < m$, which is true by the induction hypothesis. This proves Proposition 1. □

Multiplying $(DD^T)^{-1}$ from the left by the transpose difference matrix D^T has the effect of subtracting adjacent diagonals from the top-right to the bottom-left. That is, the sum of the kth diagonal

$$\sum_{i=1}^{n} b_{i+k,i}$$

for some $k \in [0, n-1]$ becomes

$$\sum_{i=1}^{n} b_{i,j-k} - \sum_{i=1}^{n} b_{i+1,j-k}.$$

Thus, the sum of the kth diagonal in the upper triangle of $\tilde{D} = D^T(DD^T)^{-1}$ becomes $\frac{T_k - T_{k-1}}{n+1}$, and the lower triangle is just the negative of this. But $T_k - T_{k-1}$ is just the kth triangular number $t_k = \frac{1}{2}k(k+1)$. So the sum of the kth diagonal of $\tilde{D}$ is

$$d_k \equiv \pm \frac{t_k}{n+1},$$

where the plus sign corresponds to the upper triangle and the minus sign to the lower triangle.

Proposition 2. The kth diagonal of $C = A\tilde{D}$ sums to

$$c_k \equiv \pm \frac{2t_k}{n+1},$$

where the plus sign corresponds to the upper triangle and the minus sign to the lower triangle.

Proof. We consider the upper and lower triangles of $N \equiv AM$, where M is an arbitrary matrix of dimension $(n-1) \times n$. Left-multiplication by A adds each row of M to the one above it, and the bottom row is cut off. In the lower triangle of M, moving an element up by one row also moves it up by one diagonal, so an equivalent view is that each diagonal is added to the one above it for $k \leq n-1$, and the element belonging to the bottom row of M is removed. As shown in (1.1), the element of the kth diagonal of the lower triangle of M that belongs to M's bottom row is just $\frac{k}{n+1}$. So, denoting the sum of the kth diagonal in the lower triangles of N and M as n_k and m_k respectively,

$$n_k = m_k + m_{k+1} - \frac{k}{n+1}.$$

But the recursion relation for the triangular sequence is $f(k) = f(k-1) + k$, so $n_k = 2m_k$. The situation for the upper triangle is exactly the same, except that the element removed from the kth diagonal of M is the one belonging to M's top row, which is also $\frac{k}{n+1}$. Thus the sum of the kth diagonal of N is twice the sum of the kth diagonal of M, except for the middle diagonal. □

Proposition 3.

$$k_\delta = \sum_{i=1}^{N-1} \sum_{j=1}^{N-1} c_{ij} \sin\big((j-i)\delta\big) = \frac{2}{N} \sum_{k=1}^{N-1} k(k+1) \sin\big((N-1-k)\delta\big). \tag{22}$$

Proof. Since the definition of the kth diagonal is $j - i = k$, the sine term is constant along each diagonal, and its sign flips exactly when C's sign flips. Thus:

$$\begin{aligned}
\sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} \sin\big((j-i)\delta\big) &= 2 \sum_{k=1}^{n} c_k \sin\big((n-k)\delta\big) \\
&= 2 \sum_{k=1}^{n} \frac{2k(k+1)/2}{n+1} \sin\big((n-k)\delta\big) \\
&= \frac{2}{n+1} \sum_{k=1}^{n} k(k+1) \sin\big((n-k)\delta\big).
\end{aligned}$$

Substituting $n = N - 1$ yields (22). $\square$

Taking a derivative of (22) with respect to δ and setting the result equal to zero gives:

$$\frac{dk_\delta}{d\delta} = \frac{1}{4N}\left[\csc^3\left(\frac{\delta}{2}\right)\right]\left[(N-2)(N+1)\cos\left(\frac{\delta}{2}\right) - N(N-1)\cos\left(\frac{3\delta}{2}\right) + 2\cos\left(\frac{\delta}{2}(1-2N)\right)\right] = 0, \quad (23)$$

which is equivalent to:

$$\left[1 + \frac{N(N-1)}{N+1}\right]\cos\delta + (N-2)\cos(\delta N) - (N+1)\left[N - 2 + \cos\big(\delta(N-1)\big)\right] = 0. \quad (24)$$

5.1 Closed-Form Approximations

The optimality condition derived above does not admit a closed-form solution, so we develop a sequence of approximations of increasing simplicity, at the cost of decreasing accuracy.

Taylor approximations. Taylor expanding the optimality condition about $\delta = 0$ and solving gives:

$$\delta_{\text{opt}} \approx \sqrt{\frac{30}{3N^2 - 3N + 2}}. \quad (25)$$

Taking an additional term in the Taylor series:

$$\delta_{\text{opt}} \approx \operatorname{Re}\sqrt{2}\sqrt{\frac{42N^2 - 42N + \sqrt{14}\,(-24N^4 + 48N^3 - 96N^2 + 72N - 34)^{\frac{1}{2}} + 28}{5N^4 - 10N^3 + 13N^2 - 8N + 3}}. \quad (26)$$

Equation (26) approximates the exact solution to within 1.6% error for all $N \geq 3$, and falls below 0.1% error for all $N \geq 40$. These expressions serve as the numerical reference against which the simpler approximations below are compared.

Continuum limit. To understand the large-N scaling, we introduce the substitution $\delta = C/(N-1)$, so that the total phase span $(N-1)\delta = C$ remains fixed as $N \to \infty$. Under this scaling, the reduced objective

$$k_\delta = \frac{2}{N}\sum_{k=1}^{N-1} k(k+1)\sin\big((N-1-k)\delta\big)$$

can be approximated by a Riemann integral. Setting $x = k/N$ so that $k \approx Nx$, and noting that $\Delta x = 1/N$, we have

$$k(k+1) \approx N^2x^2, \qquad (N-1-k)\delta \approx C(1-x),$$

and thus

$$k_\delta \approx \frac{2}{N}\cdot N\int_0^1 N^2x^2\sin\big(C(1-x)\big)\,dx = 2N\int_0^1 x^2\sin\big(C(1-x)\big)\,dx.$$

The overall factor of $2N$ does not affect the location of the optimum, so the large-N behavior is governed by

$$J(C) = \int_0^1 x^2\sin\big(C(1-x)\big)\,dx.$$

Differentiating with respect to C and setting the result to zero gives the asymptotic optimality condition:

$$\frac{dJ}{dC} = \int_0^1 x^2(1-x)\cos\big(C(1-x)\big)\,dx = 0.$$

Under the substitution $u = 1 - x$, this becomes

$$\int_0^1 (1-u)^2 u\cos(Cu)\,du = 0,$$

which is satisfied at $C \approx 4.49$, giving the asymptotic scaling

$$\delta_{\text{opt}} \sim \frac{4.49}{N-1}.$$

Simple asymptotic. A further simplification is obtained by noting that $\delta = \pi/(N-1)$ exactly maximizes the unweighted sum $\sum_{k=1}^{N-1} k \sin\big((N-1-k)\delta\big)$ and remains close to optimal for the full weighted objective k_δ. We therefore adopt

$$\delta_{\text{opt}} \approx \frac{\pi}{N-1} \tag{27}$$

as a simple alternative to (25) and (26). This expression underestimates the true optimum by approximately 30%, but has the advantage of a clean physical interpretation: the optimal joint phase offset is the value required for the first and last joints to be π out of phase, so that the body forms a discrete approximation to a single full period of the serpenoid curve.

6 Reduced-Order Model of Lateral Undulation

We develop a reduced-order model of serpentine locomotion by combining a representative-link approximation with time-scale separation and period averaging. Rather than solving the full multibody dynamics, we analyze the forces generated by a prescribed traveling-wave motion and derive their effect on the slower evolution of the center-of-mass velocity. Under these assumptions, the system admits a low-dimensional description that captures the leading-order mechanism of propulsion.

6.1 Setup and Friction Model

Consider a snake with N links of uniform length L and mass m undergoing planar motion with link centers $r_i = [x_i \quad y_i]^T$. Assuming motion primarily in the y-direction, the acceleration of link i under anisotropic viscous friction is:

$$\ddot{r}_i = -\frac{1}{m} R_i^T \Omega_i R_i \dot{r}_i \tag{28}$$

where $\Omega_i = \text{diag}(c_{n,i}, c_{t,i})$ and R_i is the rotation matrix associated with orientation

$$\phi_i = a \sin(\omega t + (i-1)\delta).$$

Expanding the friction term:

$$-R_i^T \Omega_i R_i = -\begin{bmatrix} c_n \cos^2\phi_i + c_t \sin^2\phi_i & (c_n - c_t)\sin\phi_i \cos\phi_i \\ (c_n - c_t)\sin\phi_i\cos\phi_i & c_n \sin^2\phi_i + c_t\cos^2\phi_i \end{bmatrix}.$$

The diagonal terms correspond to drag, while the off-diagonal terms couple lateral velocity and orientation, and are responsible for propulsion.

6.2 Assumptions and Reduction Principle

We consider the regime:

- Low period-averaged acceleration (near steady-state motion)
- Small amplitude $a \ll 1$
- Phase offset $\delta \approx \bar{\delta}$ (near optimal)

The sinusoidal form of ϕ_i is imposed by the control law, which drives the joints to track a traveling-wave trajectory. As a result, the body shape is not determined by passive dynamics, but is actively enforced up to tracking error.

The analysis therefore does not attempt to solve the full coupled dynamics. Instead, it evaluates the forces generated by this prescribed gait and determines their effect on the slower evolution of the center-of-mass motion. This is justified by two approximations.

First, under the feedback-linearized control law described in the previous section, the angular dynamics are input-dominated: the controller cancels nonlinear and constraint-induced forces, allowing the joint angles

to track the prescribed traveling-wave trajectory with small error. As a result, the angular motion is well approximated by

$$\phi_i(t) \approx \alpha \sin(\omega t + (i-1)\delta),$$

and its steady-state amplitude is determined primarily by the control input.

Second, the system exhibits a separation of time scales. The joint oscillations occur at frequency ω, while the forward velocity evolves under the cumulative effect of friction forces, which are $O(a^2)$ for small amplitude. Thus, the center-of-mass motion varies more slowly than the joint motion, and averaging over one oscillation period is justified.

Crucially, the phase offset δ is not an independent parameter, but scales with the number of links such that the total phase span across the body remains fixed:

$$(N-1)\delta = C.$$

This ensures that increasing the number of links refines the discretization of a fixed traveling-wave shape rather than altering the waveform itself.

Under this scaling, summations over links represent a discretization of a spatially extended but finite interaction region. Apparent divergences arising from factors such as $\csc(\delta/2)$ are compensated by the shrinking contribution of each link as N increases, yielding a finite and physically consistent limit.

Under these conditions, the coupling from translational dynamics back into the angular subsystem enters only at higher order in the amplitude. Consequently, we adopt a one-way coupling approximation: the imposed angular motion drives the translational dynamics, while the feedback from translation to joint motion is neglected to leading order. This yields a reduced model with hierarchical coupling, in which the steady-state amplitude $\bar{a}$ is determined by the angular control law and treated as an input to the translational dynamics.

6.3 Kinematics of a Representative Link

Let link k satisfy $\phi_k = 0$ at $t = 0$, so that

$$\phi_k = a \sin(\omega t).$$

To determine its lateral displacement, we approximate the contribution of upstream links over approximately half a wavelength. Let each link have length $\Delta s = L_{\text{body}}/N$. The lateral displacement is modeled as a sum of contributions from upstream links:

$$x_k \approx -\frac{1}{2} \sum_{i=1}^{M} \sin\big(a \sin(\omega t + i\delta)\big)\, \Delta s,$$

where

$$M = \left\lfloor \frac{\pi}{\delta} \right\rfloor.$$

For $a \ll 1$, we linearize:

$$\sin(a \sin(\cdot)) \approx a \sin(\cdot),$$

giving

$$x_k \approx -\frac{a\,\Delta s}{2} \sum_{i=1}^{M} \sin(\omega t + i\delta).$$

Using the identity

$$\sum_{i=1}^{M} \sin(\theta + i\delta) = \frac{\sin(M\delta/2)}{\sin(\delta/2)} \sin\left(\theta + \frac{(M+1)\delta}{2}\right),$$

we obtain

$$x_k \approx -\frac{a\,\Delta s}{2} \frac{\sin(M\delta/2)}{\sin(\delta/2)} \sin\left(\omega t + \frac{(M+1)\delta}{2}\right).$$

Since $M\delta = \pi + O(\delta)$ up to discretization error, we have

$$\sin(M\delta/2) = 1 + O(\delta),$$

so

$$x_k \approx -\frac{a\,\Delta s}{2}\csc\left(\frac{\delta}{2}\right)\sin\left(\omega t + \frac{\pi}{2} + \frac{\delta}{2} + O(\delta)\right).$$

Using $\sin(\theta + \pi/2) = \cos(\theta)$:

$$x_k \approx -\frac{a\,\Delta s}{2}\csc\left(\frac{\delta}{2}\right)\cos\left(\omega t + \frac{\delta}{2}\right) + O(\Delta s).$$

Differentiating:

$$\dot{x}_k \approx \frac{a\,\Delta s\,\omega}{2}\csc\left(\frac{\delta}{2}\right)\sin\left(\omega t + \frac{\delta}{2}\right).$$

Finally, noting that

$$\Delta s = \frac{L_{\text{body}}}{N}, \quad \delta \sim \frac{C}{N},$$

we have

$$\Delta s\,\csc\left(\frac{\delta}{2}\right) \sim \frac{L_{\text{body}}}{N}\cdot\frac{2}{\delta} \sim \frac{2L_{\text{body}}}{C},$$

which is $O(1)$. Thus the apparent divergence cancels, and the lateral velocity remains finite.

6.4 Propulsive Force Generation

The y-component of the friction force is

$$F_{ky} = -\left[(c_n - c_t)\sin\phi_k\cos\phi_k\,\dot{x}_k + (c_n\sin^2\phi_k + c_t\cos^2\phi_k)\dot{y}_k\right].$$

The first term produces propulsion. Using small-angle approximations,

$$\sin\phi_k \approx a\sin(\omega t), \quad \cos\phi_k \approx 1,$$

we obtain

$$F_{ky,\text{prop}} \approx -(c_n - c_t)\,a\sin(\omega t)\,\dot{x}_k.$$

Substituting the expression for $\dot{x}_k$:

$$F_{ky,\text{prop}} \approx -\frac{a^2\omega\,\Delta s}{2}(c_n - c_t)\csc\left(\frac{\delta}{2}\right)\sin(\omega t)\sin\left(\omega t + \frac{\delta}{2}\right).$$

Averaging over one period:

$$\left\langle \sin(\omega t)\sin\left(\omega t + \frac{\delta}{2}\right)\right\rangle = \frac{1}{2}\cos\left(\frac{\delta}{2}\right),$$

gives

$$\langle F_{ky,\text{prop}}\rangle \approx -\frac{a^2\omega\,\Delta s}{4}(c_n - c_t)\csc\left(\frac{\delta}{2}\right)\cos\left(\frac{\delta}{2}\right).$$

Using the scaling

$$\Delta s\,\csc\left(\frac{\delta}{2}\right) \sim \frac{2L_{\text{body}}}{C},$$

we obtain

$$\langle F_{ky,\text{prop}} \rangle \approx -\frac{a^2\omega}{2}\frac{L_{\text{body}}}{C}(c_n - c_t)\cos\left(\frac{\delta}{2}\right).$$

Thus, the propulsive force per link has the form

$$\langle F_{ky,\text{prop}} \rangle \propto a^2\omega\cos\left(\frac{\delta}{2}\right),$$

with a prefactor independent of N under the fixed-waveform scaling.

6.5 Steady-State Amplitude of the Angular Dynamics

The angular dynamics of each link under feedback linearization and PD control are given by

$$I\ddot{\phi} + c\dot{\phi} + b\phi = k_p\alpha\sin(\omega t) + k_d\alpha\omega\cos(\omega t), \tag{29}$$

where $b = k_p$ and the effective damping is

$$c = f_v + k_d, \tag{30}$$

with f_v denoting viscous friction due to rotation.

The viscous rotational friction coefficient f_v is obtained by approximating each half-link as a continuous rod of length $L/2$ and taking the limit of n discrete masses distributed along it. This yields

$$f_v = \lim_{n\to\infty}\frac{2c_n(L/2)^2}{n}\sum_{i=0}^{n}\left(\frac{i}{n}\right)^2. \tag{31}$$

Since

$$\frac{1}{n}\sum_{i=0}^{n}\left(\frac{i}{n}\right)^2 \to \int_0^1 x^2\,dx = \frac{1}{3},$$

we obtain

$$f_v = \frac{2}{3}c_n\left(\frac{L}{2}\right)^2. \tag{32}$$

The steady-state amplitude of this second-order system, driven by a sinusoidal input of magnitude $\alpha\sqrt{k_p^2 + k_d^2\omega^2}$, is:

$$\bar{a} = \frac{\alpha\sqrt{k_p^2 + k_d^2\omega^2}}{\sqrt{(k_p - I\omega^2)^2 + ((f_v + k_d)\omega)^2}}. \tag{33}$$

This expression reflects the combined effects of inertia, stiffness, and both physical and control-induced damping. In the high-gain regime $k_d \gg f_v$, the damping is dominated by the derivative gain and the expression simplifies accordingly.

6.6 Reduced Dynamical Equation via Period Averaging

The translational dynamics are driven by frictional forces acting on each link. The center-of-mass (COM) acceleration is obtained by summing these forces over all links and dividing by the total mass.

From the previous section, the period-averaged propulsive force acting on a representative link has the form

$$\langle F_{ky,\text{prop}} \rangle \propto \bar{a}^2\omega\cos\left(\frac{\delta}{2}\right).$$

As justified in the Assumptions section, we approximate the total force using a representative-link contribution.

Summing over all N links yields the total propulsive force

$$F_{\text{prop}} = N\langle F_{ky,\text{prop}}\rangle.$$

The total mass is $M = Nm$, so the center-of-mass acceleration is

$$\dot{Y} = \frac{F_{\text{prop}}}{M} = \frac{N\langle F_{ky,\text{prop}}\rangle}{Nm} = \frac{\langle F_{ky,\text{prop}}\rangle}{m}.$$

Thus, the factors of N cancel, and the COM acceleration inherits the same scaling as the per-link force divided by mass. We therefore write

$$\bar{f}_x = C_1\bar{a}^2\omega\cos\left(\frac{\delta}{2}\right),$$

where

$$C_1 \propto \frac{L_{\text{body}}}{C}\frac{(c_n - c_t)}{m}$$

is independent of N under the fixed-waveform scaling.

The drag force acting on each link is

$$F_{ky,\text{drag}} = -\left(c_n\sin^2\phi_k + c_t\cos^2\phi_k\right)\dot{y}_k.$$

Averaging over one oscillation period yields

$$\left\langle c_n\sin^2\phi_k + c_t\cos^2\phi_k\right\rangle \approx c_t + \tfrac{1}{2}(c_n - c_t)\bar{a}^2.$$

Summing over links and dividing by total mass produces the same structure, so we write

$$\bar{f}_y = c_t + C_2\bar{a}^2,$$

where C_2 absorbs constant factors.

The reduced dynamical model is therefore

$$\dot{Y} = \bar{f}_x - \bar{f}_y Y,$$

with

$$\bar{f}_x = C_1\bar{a}^2\omega\cos\left(\frac{\delta}{2}\right), \quad \bar{f}_y = c_t + C_2\bar{a}^2.$$

The steady-state velocity is obtained by setting $\dot{Y} = 0$:

$$Y_{\text{ss}} = \frac{C_1\bar{a}^2\omega\cos\left(\frac{\delta}{2}\right)}{c_t + C_2\bar{a}^2}.$$

For small amplitudes $\bar{a} \ll 1$, such that

$$C_2\bar{a}^2 \ll c_t,$$

the amplitude-dependent correction to drag is negligible relative to the tangential drag. In this regime, the steady-state velocity simplifies to

$$Y_{\text{ss}} \approx \frac{C_1}{c_t}\bar{a}^2\omega\cos\left(\frac{\delta}{2}\right).$$

6.7 Scaling with Number of Links

The reduced model allows a direct comparison of locomotion efficiency as the number of links varies. Using the phase offset,

$$\delta_{\text{opt}} \sim \frac{\pi}{N-1}, \quad \text{as derived in Section ??},$$

we evaluate the steady-state velocity in two limiting cases.

For $N = 3$ (two joints), we have $\delta = \pi/2$, giving

$$Y_{\text{ss}}^{(N=3)} \propto \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}.$$

In the limit $N \to \infty$, $\delta \to 0$, and thus

$$Y_{\text{ss}}^{(\infty)} \propto \cos(0) = 1.$$

Therefore, the ratio of steady-state velocities is

$$\frac{Y_{\text{ss}}^{(\infty)}}{Y_{\text{ss}}^{(N=3)}} = \sqrt{2}.$$

This comparison assumes that the geometric constant C_1 is the same in both cases. This is justified because the factor $\Delta s \csc(\delta/2)$ remains finite under the scaling $\delta \sim 1/N$, and therefore C_1 does not depend on N when the waveform is held fixed.

Crucially, both cases correspond to the same total phase span across the body,

$$C = (N-1)\delta = \pi.$$

For $N = 3$, this gives $C = 2 \cdot (\pi/2) = \pi$, and in the large-$N$ limit, the asymptotic scaling $\delta_{\text{opt}} \sim \pi/(N-1)$ also yields $(N-1)\delta = \pi$. Thus, the comparison is performed within a single fixed-waveform family, ensuring that the constants appearing in the reduced model are consistent across both cases.

This shows that increasing the number of links yields a finite improvement in propulsion efficiency, with diminishing returns as N becomes large. Additional links refine the spatial discretization of the waveform rather than fundamentally altering the propulsion mechanism.

6.8 Interpretation

Forward motion arises from a phase offset between lateral velocity and body orientation. If these signals are in phase, the motion is reciprocal over a cycle and produces no net displacement. A phase offset—induced by the traveling-wave structure—breaks this symmetry, and in the presence of anisotropic friction generates net propulsion.

The reduced model yields several scaling relationships. The steady-state velocity satisfies

$$Y_{\text{ss}} \propto \bar{a}^2 \omega \cos\left(\frac{\delta}{2}\right),$$

exhibiting quadratic dependence on oscillation amplitude, linear dependence on driving frequency, and requiring anisotropic friction ($c_n > c_t$) for propulsion.

The dependence on phase offset arises from the interaction between lateral velocity and body orientation. The apparent geometric amplification associated with $\csc(\delta/2)$ reflects the accumulation of contributions across multiple links, but when properly normalized by link length, this factor remains finite and does not alter the leading-order phase dependence.

The reduced model also enables analysis of the optimal driving frequency. Since the steady-state velocity scales as $Y_{\text{ss}} \propto \bar{a}^2(\omega)\,\omega$, maximizing propulsion requires balancing the frequency-dependent amplitude response of the joint dynamics with the linear increase in propulsion with ω. Using the expression for $\bar{a}$, this

leads to an implicit optimality condition for ω, which can be solved numerically. In the lightly damped regime, the optimal frequency occurs near the natural frequency of the joint dynamics, $\omega_{\text{opt}} \approx \sqrt{k_p/I}$, with damping shifting the optimum to lower values. This highlights that efficient locomotion arises from tuning the driving frequency to the mechanical response of the system.

The model also provides a characteristic convergence time scale. Linearizing the dynamics yields

$$\tau \sim \frac{1}{c_t},$$

to leading order, indicating that the dominant time scale is set by tangential drag, with amplitude-dependent corrections entering at higher order.

Finally, because the derivation assumes a fixed total phase span across the body, decreasing δ corresponds to increasing the number of links while preserving the overall waveform. In this regime, the model remains well-behaved and does not exhibit unphysical divergence.

7 Path Following

To evaluate the feedback-linearized controller in a control setting, we consider a path-following task in which the snake robot tracks a sequence of waypoints by modulating the gait offset parameter ϕ_0.

Under the traveling-wave control law (Equation (8)), the robot moves in a straight line when $\phi_0 = 0$ and follows a trajectory with approximately constant curvature when ϕ_0 is held constant [Sato et al., 2002]. Thus, steering can be achieved by dynamically adjusting ϕ_0.

A common approach [Liljebäck et al., 2013] defines the reference offset as:

$$\phi_{0,\text{ref}} = A_h(h_{\text{ref}} - h), \tag{34}$$

where h is the average heading of the robot,

$$h = \frac{1}{N-1} \sum_{i=1}^{N-1} \theta_i,$$

and h_{ref} is the desired heading direction.

For waypoint tracking, the desired heading can be computed geometrically as:

$$h_{\text{ref}} = \arctan\left(\frac{P_Y}{\Delta - P_X}\right), \tag{35}$$

where (P_X, P_Y) represents the relative position of the target waypoint.

To improve robustness, we instead formulate heading control in terms of cross-track error. Let $K_{i,j}$ denote the position of the robot and E_j the j-th waypoint. The cross-track error is defined as:

$$P_{X,j} = \pm\sqrt{(K_{i,j,x} - E_{j,x})^2 + (K_{i,j,y} - E_{j,y})^2}, \tag{36}$$

where the sign is determined by the relative orientation between the robot velocity and the waypoint direction:

$$P_{X,j} = -\text{sgn}(\theta_{\text{rel}} - \theta_{\text{wp}})\, P_{X,j}.$$

Here,

$$\theta_{\text{rel}} = \arctan 2(v_{i,j,y}, v_{i,j,x}), \quad \theta_{\text{wp}} = \arctan 2(w_{j,y}, w_{j,x}), \tag{37}$$

represent the robot's instantaneous heading and the direction to the waypoint, respectively.

The desired heading is then generated using a PID controller:

$$h_{\mathrm{ref},j} = K_P P_{X,j} + K_I \int P_{X,j}\, dt + K_D \dot{P}_{X,j}. \tag{38}$$

To improve performance across different motion regimes, the proportional gain may be adapted based on the angle of attack ϕ_{AoA}:

$$K_P = p_1\big(\arctan 2(p_2 \phi_{\mathrm{AoA}}) + p_3\big). \tag{39}$$

This control strategy modulates the gait offset ϕ_0 to steer the robot while the feedback-linearized controller ensures accurate tracking of the underlying traveling-wave motion.

8 Experiments

The reduced model was compared to a full multibody simulation using MATLAB `ode45` over a 30-second interval, with the system initialized from rest.

At fixed parameter settings, the reduced model reproduces the key qualitative features of the multibody dynamics. In particular, both models exhibit a transient oscillatory regime in the center-of-mass velocity and convergence to a steady-state forward speed, as shown in Figure 1. The dependence of both oscillation amplitude and steady-state velocity on the driving frequency ω is also captured. Across a range of frequencies, higher ω leads to smaller oscillation amplitudes and higher steady-state speeds, a trend that is evident in Figure 1.

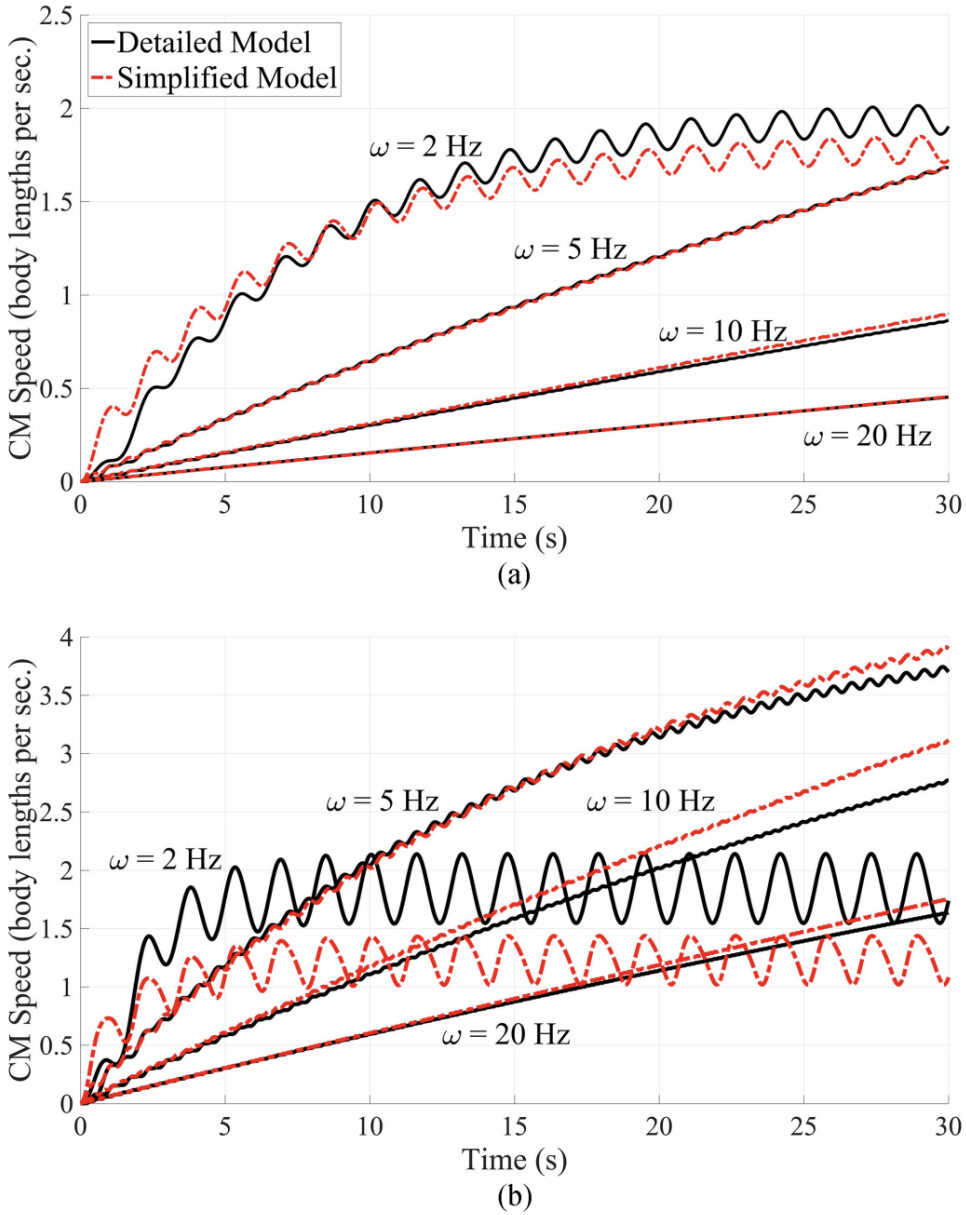

Figure 1: Comparison of the full multibody dynamic model (black solid line) with the single-body model (red dotted-line) at various oscillation frequencies for (a) $\alpha = 0.5$ rad and (b) $\alpha = 1$ rad.

Quantitatively, the agreement depends on the operating regime. For sufficiently large ω, the reduced model achieves $< 1\%$ error in steady-state amplitude for $\omega > 10$ Hz, while at lower frequencies the error increases to $\sim 40\%$ at $\omega = 2$ Hz. This degradation is consistent with the increasing discrepancy between the two models visible at lower frequencies in Figure 1.

These discrepancies follow directly from the assumptions underlying the reduced model. The derivation relies on (i) small-amplitude motion, (ii) near steady-state dynamics, and (iii) a separation of time scales between fast joint oscillations and the slower evolution of the center-of-mass velocity. At low frequencies, this time-scale separation breaks down, and the forward velocity $Y(t)$ no longer evolves slowly relative to the oscillatory forcing terms, invalidating the averaging argument. Similarly, at larger amplitudes, the small-angle approximations used in deriving $\dot{x}_k$ and the propulsive force no longer hold, and inter-link coupling

effects become significant.

In addition, the reduced model collapses the full multibody dynamics into a small number of effective coefficients (e.g., C_1, C_2), which absorb the effects of constraint forces and coupling. While this captures the dominant behavior when motion is smooth and coherent, it becomes sensitive to parameter choices when these conditions are violated.

The dependence of steady-state speed on frequency and body length is shown in Figure 2. Increasing frequency leads to higher steady-state speeds, while longer bodies produce greater propulsion due to increased phase-coherent interactions along the chain.

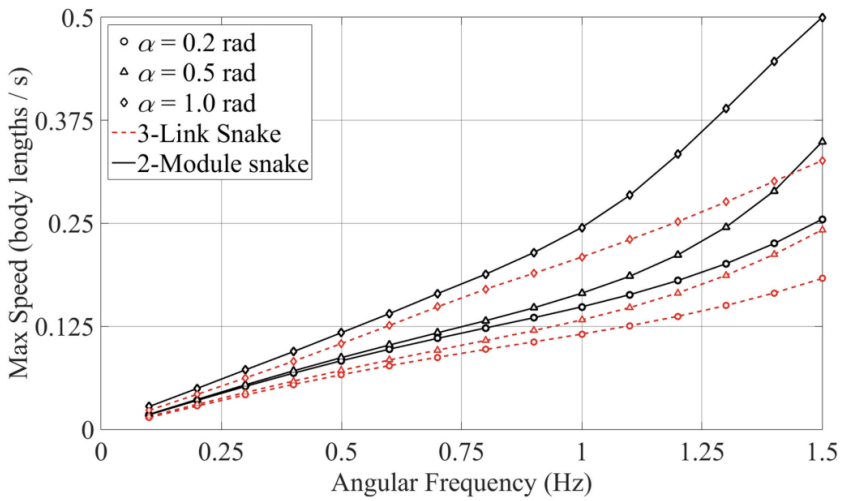

Figure 2: Simulated steady-state speeds for snake robots with different body lengths as the driving amplitude and frequency are varied.

Overall, the reduced model should be interpreted as capturing the dominant dynamical structure of serpentine locomotion rather than providing uniformly accurate predictions. Within its regime of validity, it offers a compact description of how phase relationships and anisotropic friction generate propulsion.

Finally, Figure 3 illustrates a path-following task using the proposed control strategy. By modulating the offset parameter ϕ_0, the snake robot is able to track a sequence of waypoints and execute complex planar trajectories.

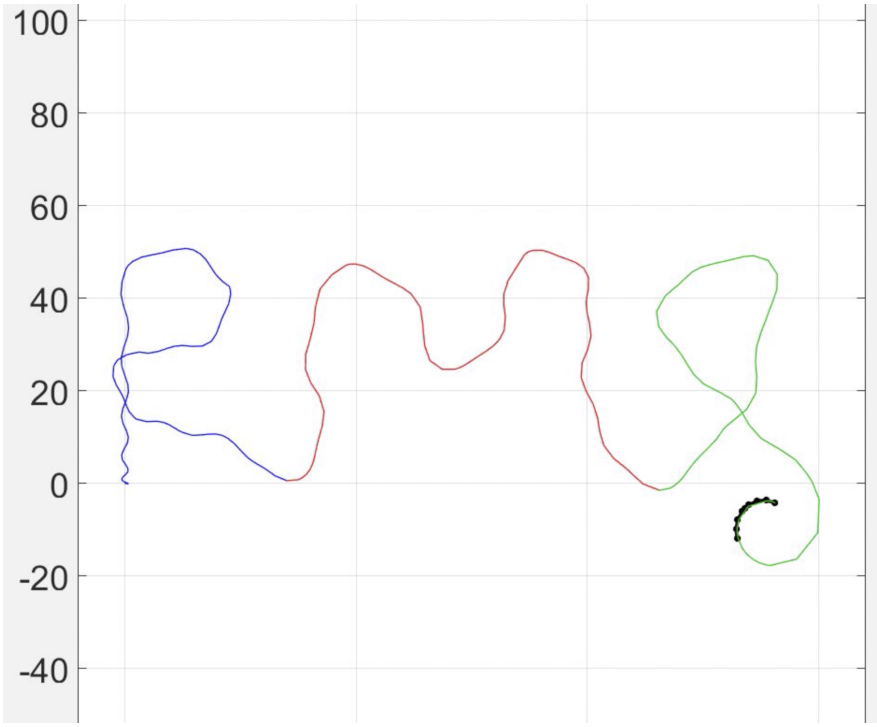

Figure 3: Simulated path-following control using waypoint guidance.

9 Conclusion

We revisited classical models of serpentine locomotion and showed that, despite the apparent complexity of multibody dynamics, the system admits a simple underlying structure. In particular, we demonstrated that locomotion can be understood as a low-dimensional traveling-wave process, in which both kinematics and dynamics are governed by a small set of parameters.

By analyzing the interaction between body orientation, lateral motion, and anisotropic friction, we derived a reduced dynamical model in which forward velocity is governed by a single first-order equation. This formulation makes explicit the mechanism of propulsion as a phase offset between lateral velocity and body orientation, induced by the traveling-wave structure of the gait.

We further showed that the optimal joint phase offset can be approximated in closed form and is determined by a structured weighting of phase interactions along the body, rather than a purely geometric phase partition. This provides a principled connection between gait parameters and locomotion performance.

Together, these results unify kinematic descriptions, dynamic models, and control strategies within a common low-dimensional framework. Beyond computational efficiency, this perspective provides an interpretable account of how coordinated body motion and anisotropic friction generate propulsion, and may inform the design of more effective control policies and robot morphologies.

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